

PROJECT REPORT

FOODBRIDGE

Food Management & Donation System

"Where Extra Food Finds a Purpose."

Programming in Java – Project Report

Submitted By: HIMANI SINGH THAKUR

Registration / Roll No:- 25BAI11283

Course: Programming in java

Branch: CSE(AIML)

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1. Introduction

FoodBridge is a Java-based food management and donation system developed to provide a simple way to record surplus food and connect it with NGOs that can receive it. Food wastage can occur in homes, restaurants, cafeterias, institutions, events, and other places where food is prepared in excess. FoodBridge focuses on organizing this information and supporting a basic donation workflow.

The project is implemented as a menu-driven Core Java application. It demonstrates object-oriented programming concepts, classes and objects, methods, conditional statements, loops, collections, and user input through a practical real-world use case.

2. Problem Statement

Surplus food is often available while people and communities may need food assistance. Without a simple system, information about available food, quantities, expiry dates, donation status, and suitable NGOs can be difficult to organize. Manual tracking may also make it difficult to know which food has already been donated.

FoodBridge addresses this problem by allowing users to add surplus food, explore and search food records, maintain an NGO directory, donate available food to an NGO, and view donation history and basic impact information.

3. Functional Requirements

1. **Add Surplus Food:** The user can enter food name, quantity, unit, food type, and expiry date
2. **Explore Food:** The system displays food records available in the application.
3. **Search Food:** The user can search for a food item using relevant information.
4. **Donate Food:** The user can select an available food item and donate it to a registered NGO.
5. **Update Food Status:** After successful donation, the selected food is marked as Donated and the receiving NGO is recorded.
6. **NGO Directory:** The system displays registered NGOs and their relevant information.
7. **Add NGO:** The user can add a new NGO to the directory.
8. **Donation History:** The system displays food items that have already been donated.
9. **Impact Dashboard:** The system displays basic counts such as total food, available food, donated food, and registered NGOs.

4. Non-functional Requirements

- Usability – The menu-driven interface should be simple and easy to understand.
- Performance – Operations on the small in-memory data collection should execute quickly.
- Maintainability – Food and NGO operations are divided into separate classes.
- Reliability – The application should maintain a consistent donation status during a session.
- Portability – The project should run on a computer with a compatible Java Development Kit.
- Readability – Classes, methods, and variables should use meaningful names and clear logic.

5. System Architecture

FoodBridge follows a simple object-oriented architecture suitable for a Core Java console application. Main.java controls user interaction and the main application flow. FoodManager handles food-related

operations, while NGOManager handles NGO-related operations. Food and NGO represent the main data objects.

High-level architecture:

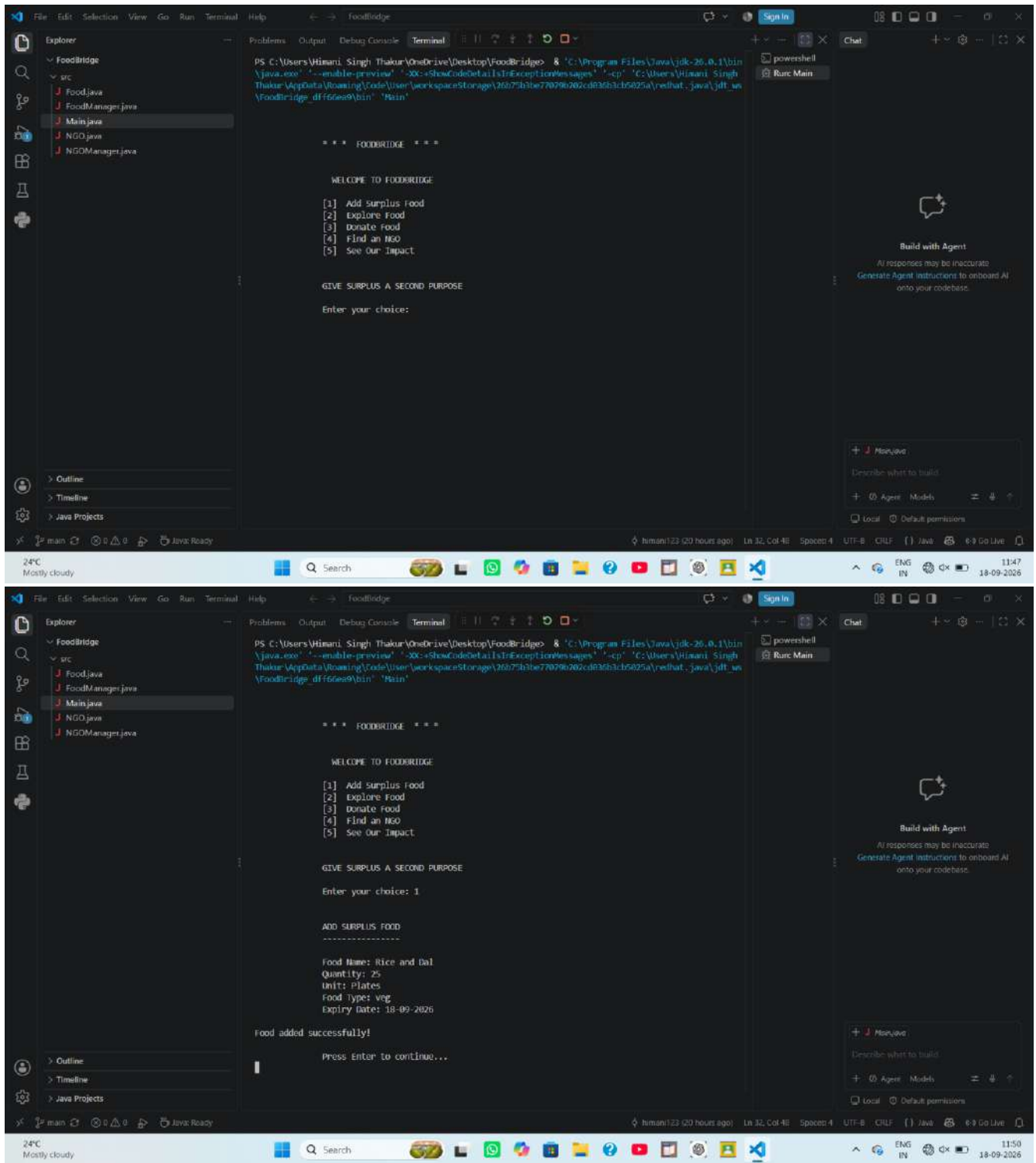
User → Main.java → FoodManager / NGOManager → Food / NGO Objects

The current implementation uses ArrayList for temporary in-memory storage. No database is used in the current version, so records added during a session are not persistent after the program is closed.

6. Design Diagrams

6.1 Use Case Diagram

The Use Case Diagram represents the interaction between the user and the main FoodBridge functions. The user can add food, explore food, search food, manage NGOs, donate food, view donation history, and view the impact dashboard.



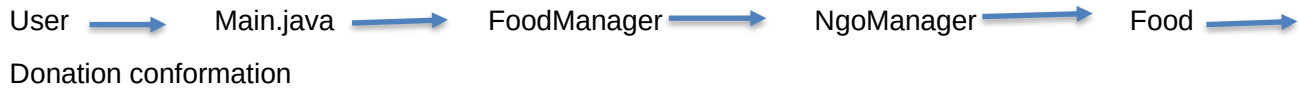
6.2 Workflow Diagram

The workflow describes the general sequence followed by the application from startup to completion.

Start → Display Main Menu → Select Feature → Enter / Select Data → Process Request → Display Result → Return to Menu → Exit

6.3 Sequence Diagram

A typical donation sequence is: the user selects Donate Food; Main.java receives the selection; FoodManager provides available food; NGOManager provides registered NGOs; the selected Food object is updated with the NGO and Donated status; and the result is displayed to the user.



6.4 Class/Component Diagram

The main components are Main, FoodManager, NGOManager, Food, and NGO. Main coordinates the application. FoodManager manages Food objects, and NGOManager manages NGO objects.

6.5 ER Diagram (if storage is used)

An ER Diagram is not applicable to the current implementation because the project does not use a database or persistent storage. Food and NGO information is stored temporarily using Java ArrayList collections.

7. Design Decisions & Rationale

Core Java: Java was selected because the project demonstrates programming fundamentals and object-oriented concepts.

Object-Oriented Classes: Food and NGO are represented as separate classes so their data and behavior can be managed clearly.

Manager Classes: FoodManager and NGOManager separate management operations from the main application flow.

ArrayList: ArrayList provides simple temporary storage for multiple Food and NGO objects.

Menu-driven Interface: A console-based menu keeps the project focused on Java programming and makes testing straightforward.

Food Status: Available and Donated states make it possible to distinguish food that can still be donated from food that has already been donated.

Pre-added NGOs: Initial NGO records make the donation feature easy to demonstrate during testing.

8. Implementation Details

8.1 Project Structure

```
FoodBridge
├── src
│   ├── Main.java
│   ├── Food.java
│   ├── FoodManager.java
│   ├── NGO.java
│   └── NGOManager.java
```

8.2 File Description

Main.java: Controls the application flow, displays menus, reads input, and calls the required manager methods.

Food.java: Defines the Food object and stores food details, status, and donation information.

FoodManager.java: Stores and manages Food objects using ArrayList and provides food-related operations.

NGO.java: Defines the NGO object and stores NGO information.

NGOManager.java: Stores and manages NGO objects, including initial and newly added NGO records.

8.3 Technologies / Tools Used

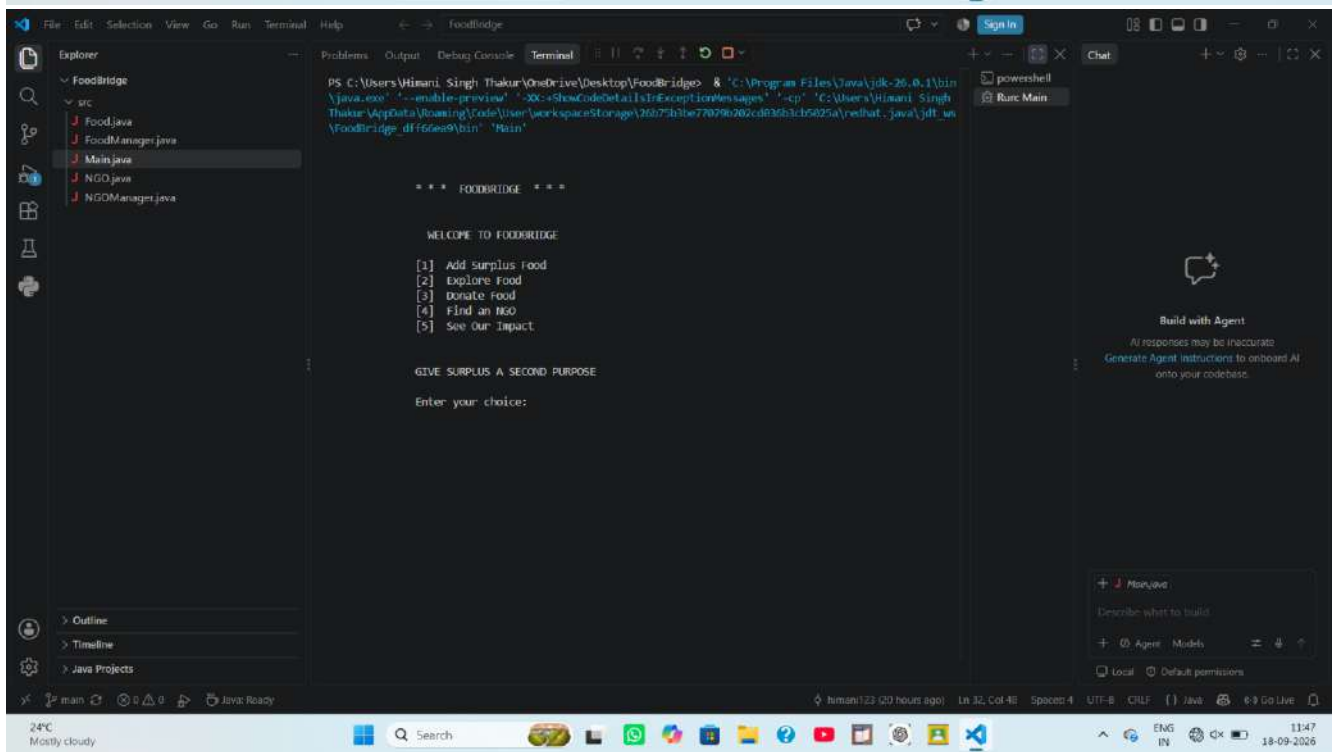
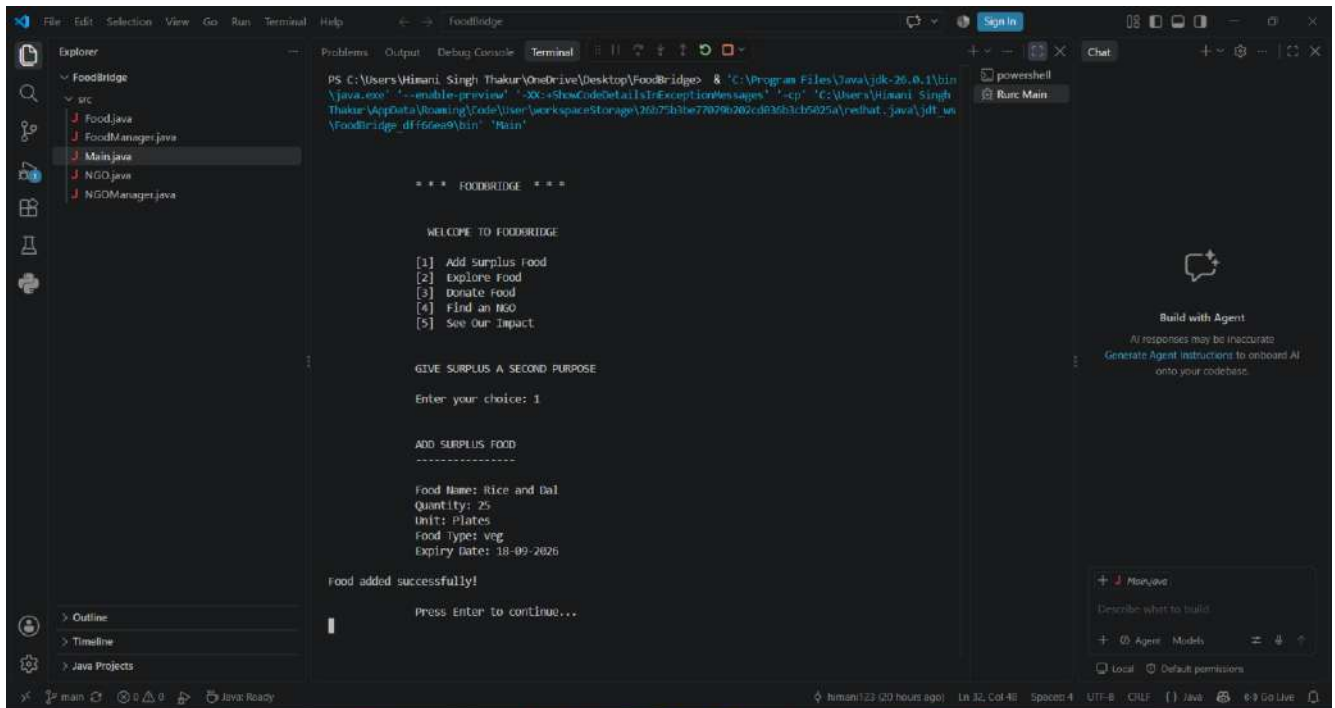
- Core Java – Main programming language.
- Object-Oriented Programming – Classes and objects for food and NGO management.
- ArrayList – Temporary in-memory collection.
- Java Development Kit (JDK) – Compilation and execution.
- Visual Studio Code – Development and code editing environment.
- Git / GitHub – Version control and repository hosting.

8.4 Data Handling

Food and NGO objects are stored in ArrayList collections. A newly added food item is initially available. During donation, the selected Food object is updated with the receiving NGO and its status is changed to Donated.

9. Screenshots / Results

Screenshots of the actual working project should be inserted in this section. The recommended order shows the complete workflow of FoodBridge.



File Edit Selection View Go Run Terminal Help

FoodBridge

Explorer

- FoodBridge
 - src
 - Food.java
 - FoodManager.java
 - Main.java
 - NGO.java
 - NGOManager.java

Problems Output Debug Console Terminal

```
FOODBRIDGE HOME
[1] Add Surplus Food
[2] Explore Food
[3] Search Food
[4] Donate Food
[5] Donation History
[6] NGO Directory
[7] Add NGO
[8] Impact Dashboard
[9] Back to Welcome
[0] Exit

Enter your choice: 6

NGO DIRECTORY
-----
[1] Robin Hood Army
    Location : Bhopal
    Contact : Contact to be verified
    Accepts : Cooked Food, Packaged Food

[2] Feeding India
    Location : Bhopal
    Contact : Contact to be verified
    Accepts : Cooked Food, Packaged Food

[3] Akshaya Patra Foundation
    Location : Bhopal
    Contact : Contact to be verified
    Accepts : Cooked Food

[4] Annamrita Foundation
    Location : Bhopal
    Contact : Contact to be verified
    Accepts : Cooked Food
```

Chat

Build with Agent

AI responses may be inaccurate. Generate Agent instructions to onboard AI onto your codebase.

main 24°C Mostly cloudy

himani123 (20 hours ago) Ln 32, Col 48 Spaces: 4 UTF-8 CRLF Java 6-8 Go Live

File Edit Selection View Go Run Terminal Help

FoodBridge

Explorer

- FoodBridge
 - src
 - Food.java
 - FoodManager.java
 - Main.java
 - NGO.java
 - NGOManager.java

Problems Output Debug Console Terminal

```
[5] Gulab jamun
Quantity : 100 Packets
Type : veg
Expiry : 21-09-2026
Status : Available

Press Enter to continue...

FOODBRIDGE HOME
[1] Add Surplus Food
[2] Explore Food
[3] Search Food
[4] Donate Food
[5] Donation History
[6] NGO Directory
[7] Add NGO
[8] Impact Dashboard
[9] Back to Welcome
[0] Exit

Enter your choice: 3

SEARCH FOOD
-----

Enter food name: Biryani

Biryani
Quantity : 65 Plates
Type : veg
Expiry : 19-09-2026
Status : Available

Press Enter to continue...
```

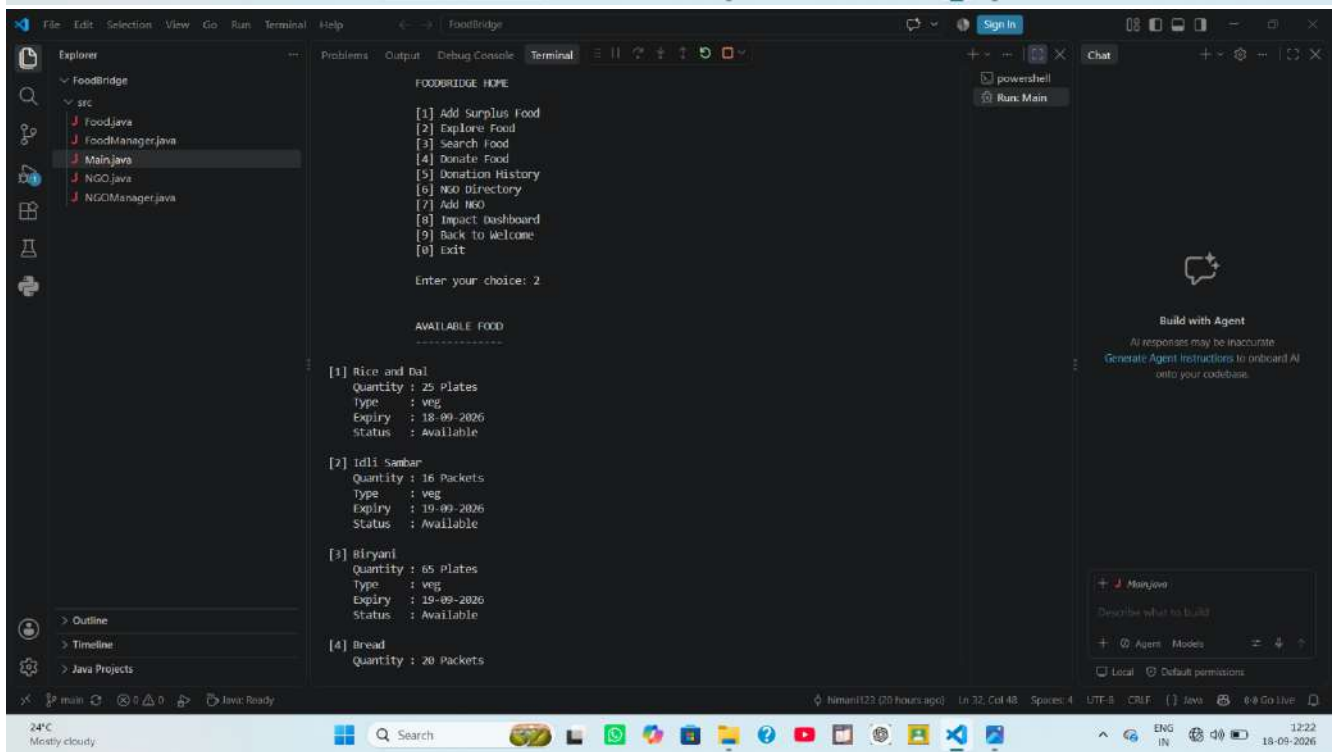
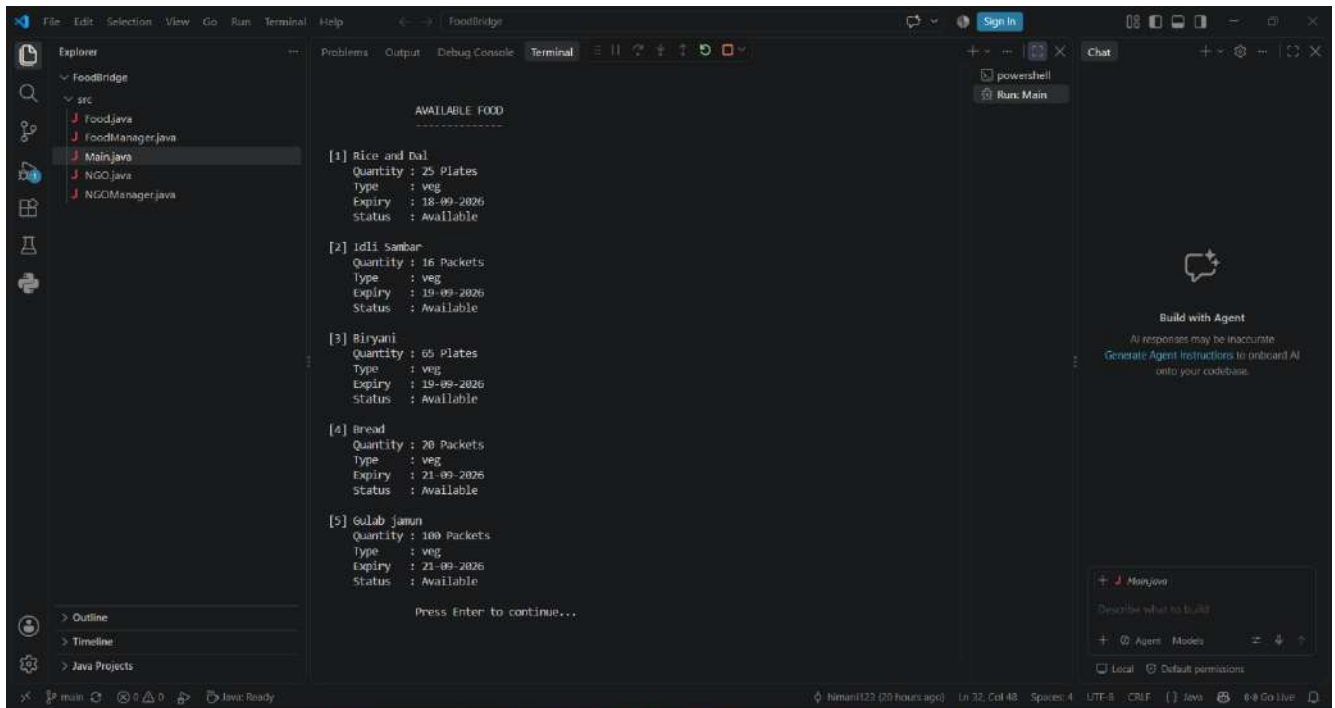
Chat

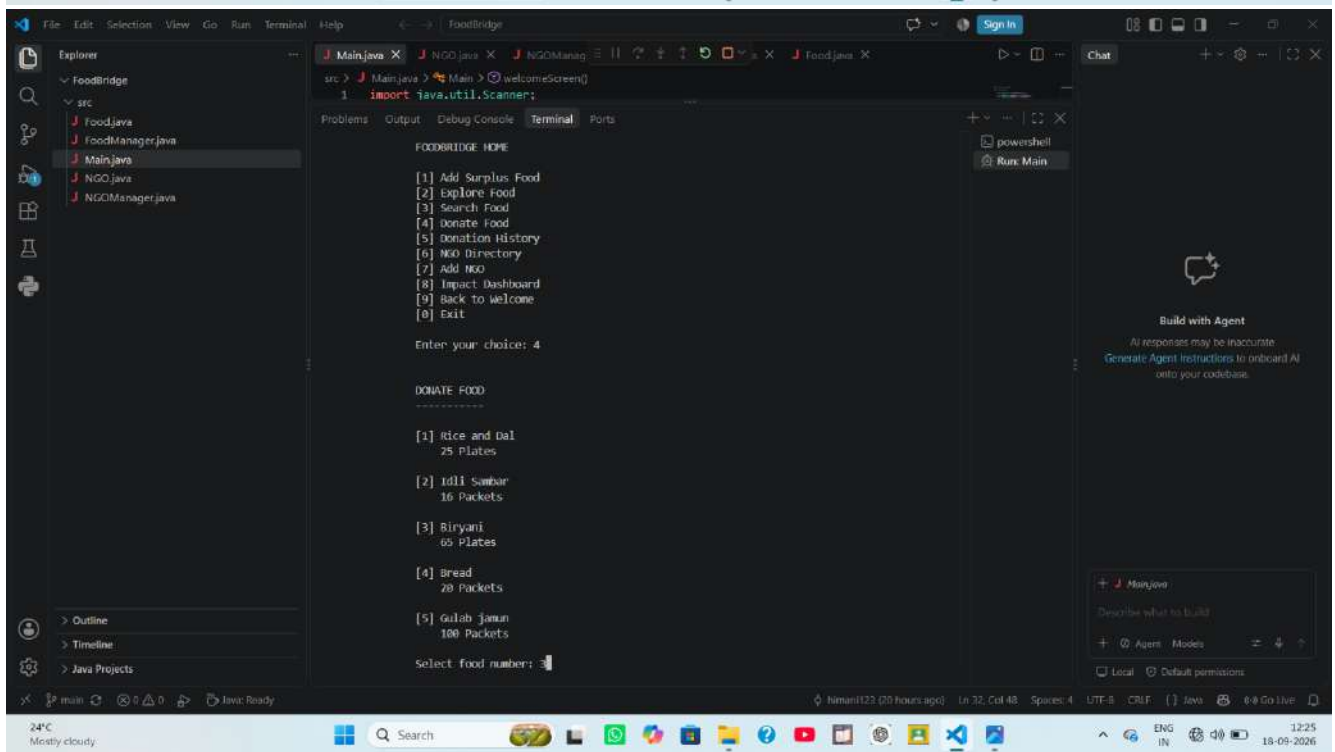
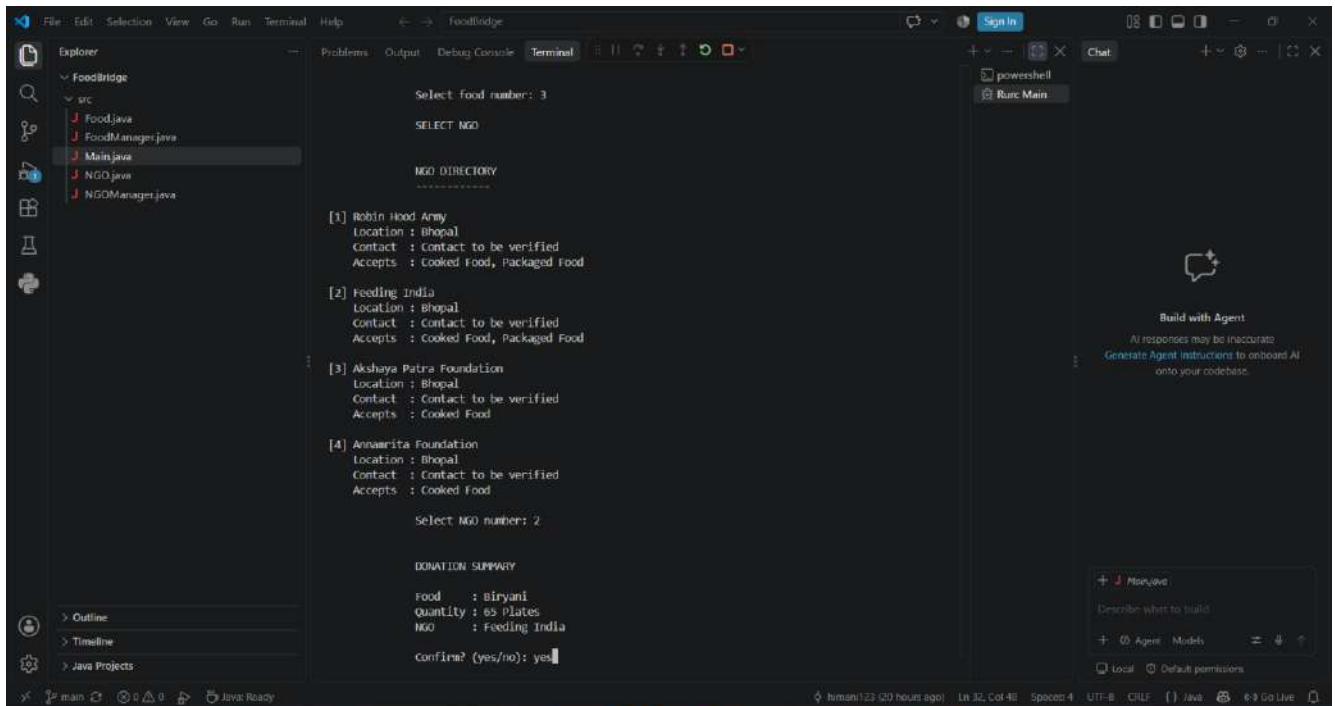
Build with Agent

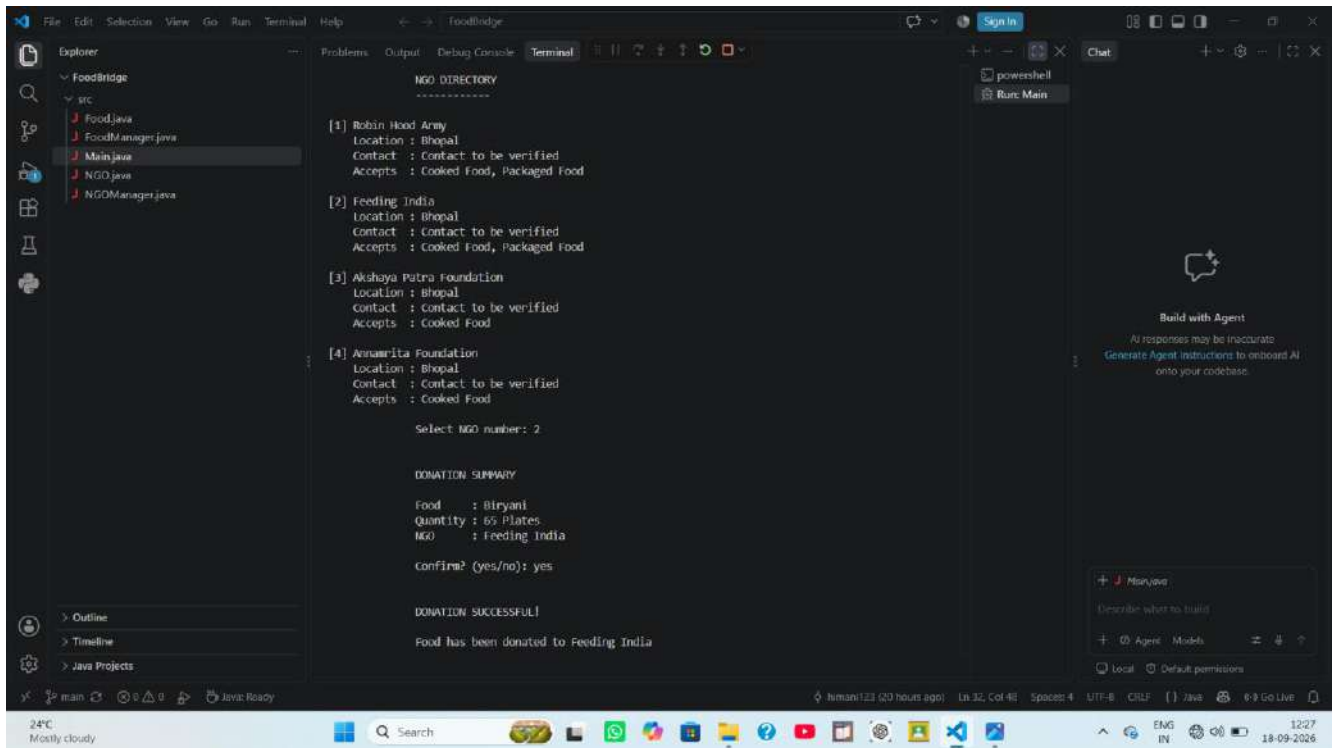
AI responses may be inaccurate. Generate Agent instructions to onboard AI onto your codebase.

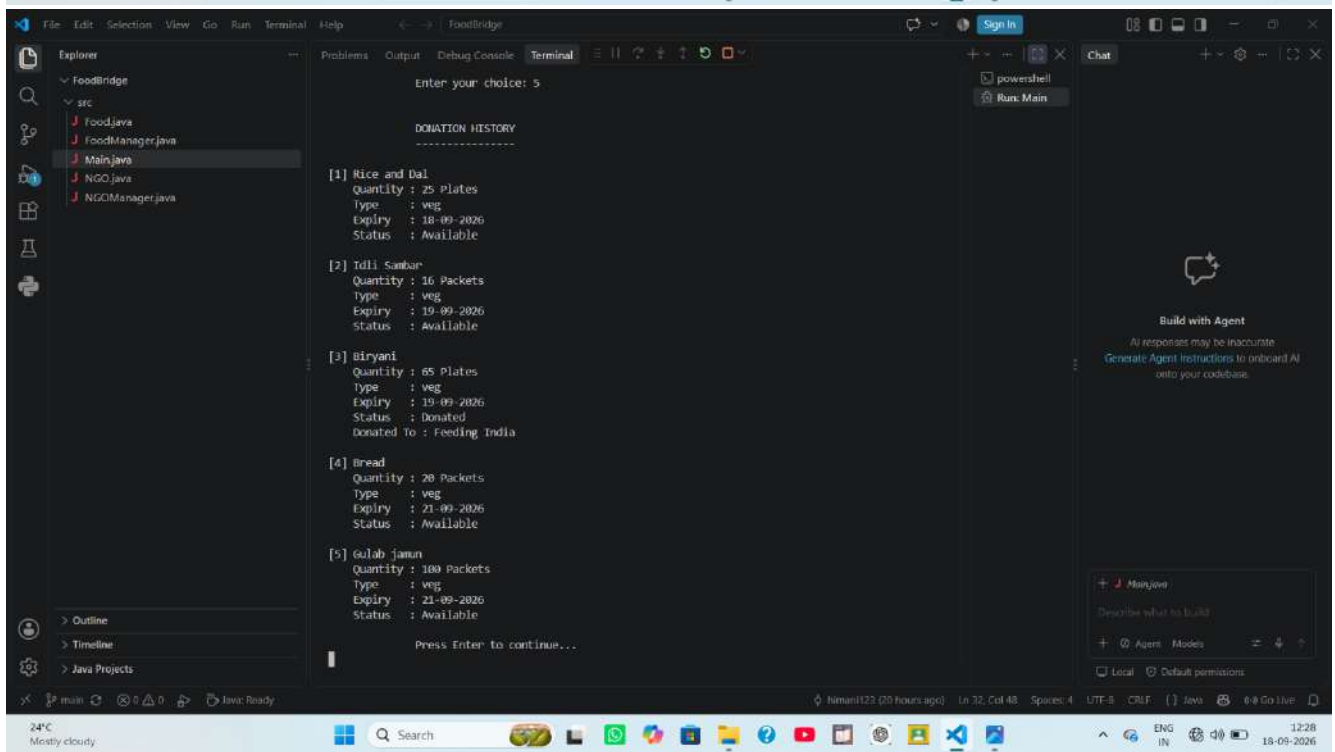
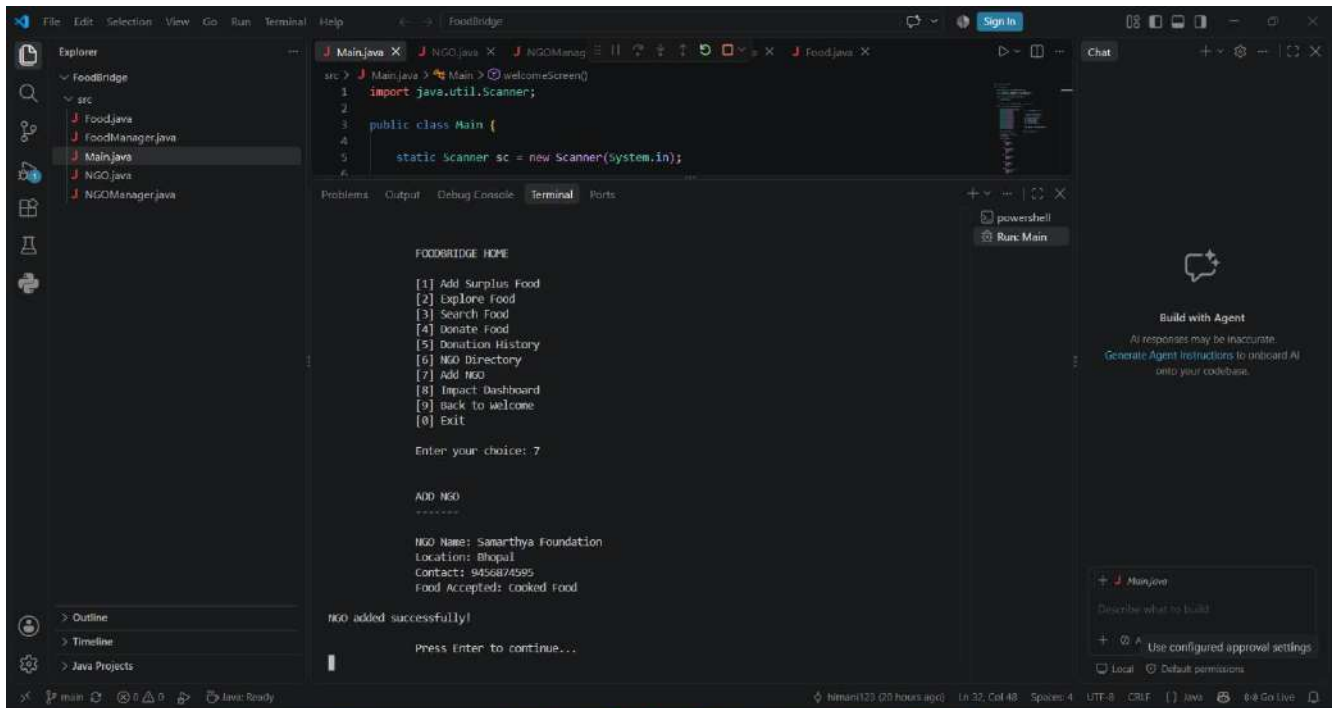
main 24°C Mostly cloudy

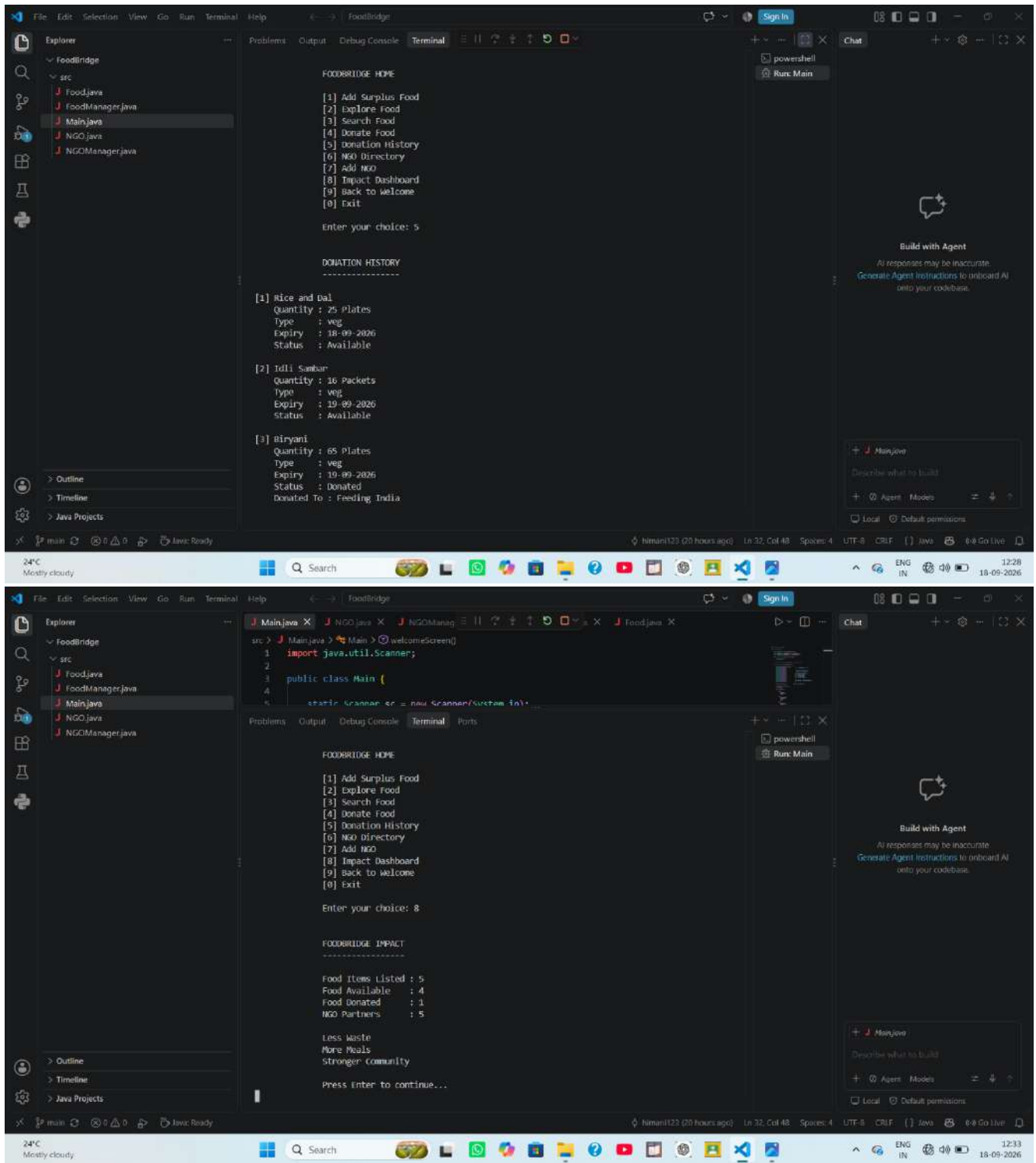
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10. Testing Approach

Testing was performed using feature-based functional test cases. Each major menu option was executed with sample inputs, and the observed output was compared with the expected behavior.

Test Case	Input / Action	Expected Result	Status
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1	Add food with valid details	Food is added and marked Available	Pass
2	Open Explore Food	Food records are displayed	Pass
3	Search an existing food item	Matching food record is displayed	Pass
4	Open NGO Directory	Registered NGOs are displayed	Pass
5	Add a new NGO	New NGO is added to the directory	Pass
6	Donate available food	Food becomes Donated and NGO is recorded	Pass
7	Open Donation History	Donated record is displayed	Pass
8	Open Impact Dashboard	Basic project counts are displayed	Pass

11. Challenges Faced

- Managing multiple Java classes and ensuring that all classes compile together.
- Handling user input for both text and numeric values.
- Connecting each menu option in Main.java with the correct manager method.
- Keeping food status consistent after a donation.
- Testing the application with multiple food and NGO records.
- Working with temporary storage because the current version does not use a database.

12. Conclusion

FoodBridge provides a simple implementation of a food management and donation workflow using Core Java. The project demonstrates how object-oriented programming and Java collections can be used to address a practical problem. The application supports food entry, exploration, search, NGO management, donation, donation history, and basic impact reporting.

The current version is intentionally simple and uses in-memory storage. Future versions can introduce persistent database storage, a web interface, authentication, expiry reminders, NGO verification, pickup tracking, and notification features.

